

AI Resume Analyzer

Introduction

The AI Resume Analyzer is a web-based application designed to help job seekers evaluate and improve their resumes by comparing resume content against job roles and industry expectations. It provides intelligent parsing of resume text, identifies key skills, detects gaps, and suggests improvements so users can present stronger, more relevant profiles to recruiters. The application leverages Python, Flask, and database storage to manage user accounts and analysis history. By supporting document upload and text extraction from common formats, it gives users a practical tool to enhance their resume quality and job readiness.

Problem Statement

Many job seekers struggle to determine whether their resumes align with the requirements of technical roles. Manual resume review is often slow, inconsistent, and inaccessible to early-career applicants. The AI Resume Analyzer addresses the following key challenges:

- Lack of clear alignment between resume content and target technical roles such as software engineer, data scientist, or web developer.
- Difficulty identifying missing or weak skills in a resume.
- Limited access to objective resume feedback for students and early-career professionals.
- Challenges in extracting meaningful insight from multiple resume formats like Word and PDF.
- Need for a tool that preserves analysis history and supports continuous improvement.

Objective

The project objective is to develop a compact and user-friendly resume review system that provides actionable feedback aligned to role-specific skill expectations. The core goals

include:

- Build a Flask-based web application with secure user login and session management.
- Support resume input through PDF and Word uploads with text extraction.
- Compare resume content against curated skill sets for key technical roles.
- Highlight strengths, gaps, and recommendations for improvement.
- Maintain analysis history so users can compare progress over time.
- Provide a clear dashboard and export-ready reporting capabilities.

Methodology

The development followed a structured methodology covering requirements, design, implementation, and validation. Key steps included:

- **Requirement Gathering:** Analyzed typical resume evaluation criteria and identified user personas that would benefit from an automated reviewer.
- **Architecture Design:** Selected Flask for server logic and SQLAlchemy for database persistence, enabling organized user and report management.
- **Skill Mapping:** Constructed predefined role skill sets for software engineer, data scientist, data analyst, web developer, machine learning engineer, and DevOps engineer.
- **Resume Parsing:** Implemented text extraction from uploaded documents and normalized content for keyword matching.
- **Analysis Logic:** Matched resume text against role-specific skills, quantified coverage, and produced clear recommendations.
- **User Interface:** Built a simple, functional frontend with templates for login, dashboard, history, and detailed report views.

Limitations

- The system uses rule-based keyword matching, so it may miss nuanced phrasing or contextual strengths in resumes.
- Document parsing can be affected by complex layouts, graphics, or non-standard formatting in uploaded files.
- Only a defined set of technical roles is supported, which limits applicability for highly specialized careers.
- The current interface is functional but not optimized for advanced user experience or design polish.
- Scaling to large numbers of users would require more robust deployment and database architecture.

Conclusion

The AI Resume Analyzer provides a working prototype for automated resume review with structured feedback, secure user access, and historical reporting. It helps candidates understand role-specific expectations and identify areas where their resumes can be strengthened. The project demonstrates strong alignment between applied Python web development and resume analytics requirements, while also highlighting future improvements for smarter NLP, expanded role coverage, and a more polished user experience. This 3-page synopsis summarizes the project direction, technical approach, and real-world usefulness of the AI Resume Analyzer.

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